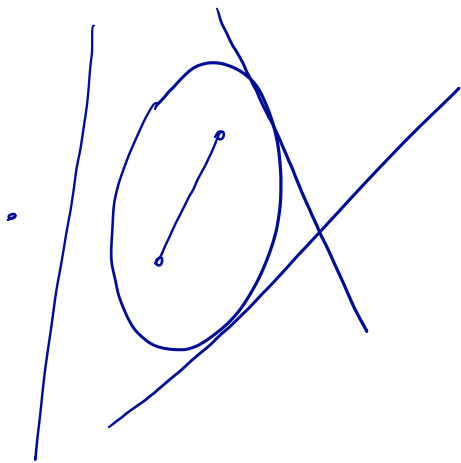
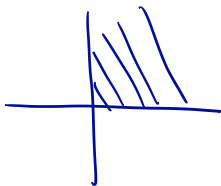


Some examples of convex sets



The positive orthant

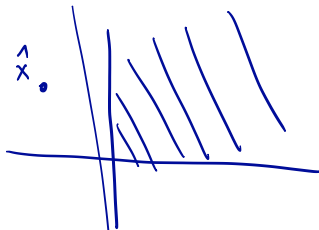


$$\mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_i \geq 0\}$$

Separating hyperplane

Given $x \in \mathbb{R}^n \setminus \mathbb{R}_+^n$

find a "separating hyperplane"



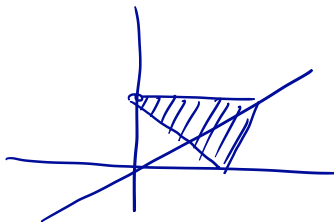
$$\hat{x} \notin \mathbb{R}_+^n \Rightarrow \exists i \text{ s.t. } \hat{x}_i < 0 \text{ - say that } i=1. \quad \hat{x} = \begin{pmatrix} -1 \\ \vdots \end{pmatrix}$$

$$s = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = e_1, \quad b = 0 \quad s^T \hat{x} < b \Rightarrow \hat{x} \in H_{s,b}^-$$

$$\forall x \in \mathbb{R}_+^n, \quad s^T x \geq 0 = b \quad \mathbb{R}_+^n \subseteq H_{s,b}^+$$

The simplex in n dimensions

$$\{x \in \mathbb{R}^n : \sum x_i = 1, x_i \geq 0\}$$



The set of moments of distributions with given support

Let $C \subseteq \mathbb{R}^n$, $\mathcal{P}_C$ = set of prob. distributions with support on C .

If $p \in \mathcal{P}_C$, $\mathbb{E}_p \left[x_i \right]$ ^{any monomial} - a moment of p .

Ex: $n=1$, $E[x]$, $E[x^2]$, $\dots$

$m_p^\alpha = \mathbb{E}_p [x^\alpha]$, α - multi-index: $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$
 $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$

$m_{p_1}^\alpha, m_{p_2}^\alpha \Rightarrow \lambda m_{p_1}^\alpha + (1-\lambda) m_{p_2}^\alpha$ is also a moment of a distribution with support on C .

Exercise: show that $p = \lambda p_1 + (1-\lambda) p_2$ is a prob. dist'n with support on C . Then: $\lambda m_{p_1}^\alpha + (1-\lambda) m_{p_2}^\alpha = \mathbb{E}_p [x^\alpha]$.

Positive semidefinite matrices

M $n \times n$ matrix $\Rightarrow$ psd if:

- a. symmetric
- b. has non-neg. e-values
- b'. $x^T M x \geq 0 \quad \forall x \in \mathbb{R}^n$.

$$M_1, M_2 \in \mathcal{S}_+^n$$

$$M = \lambda M_1 + (1-\lambda) M_2 \stackrel{?}{\in} \mathcal{S}_+^n, \quad \lambda \in [0, 1]$$

Need to check: $x^T M x \geq 0 \quad \forall x$?

$$x^T M x = x^T (\lambda M_1 + (1-\lambda) M_2) x = \underbrace{\lambda x^T M_1 x}_{\geq 0} + \underbrace{(1-\lambda) x^T M_2 x}_{\geq 0} \geq 0 \quad \checkmark$$

Positive semidefinite matrices: separating hyperplane

Suppose $\hat{M}$ is not psd. (but symmetric).

$$\hat{M} \text{ symmetric} \Rightarrow \hat{M} = \sum \hat{\lambda}_i \hat{v}_i \hat{v}_i^T \quad \hat{M} \text{ not psd} \Rightarrow \exists i \text{ s.t. } \hat{\lambda}_i < 0$$

say $\lambda_1 < 0$.

$$s = \hat{v}_1 \hat{v}_1^T, \quad b = 0$$

$$\hat{M} \in H_{s,b}^-$$

$$\begin{aligned} \langle s, \hat{M} \rangle &= \langle \hat{v}_1 \hat{v}_1^T, \hat{M} \rangle = \hat{v}_1^T \hat{M} \hat{v}_1 \\ &= \hat{v}_1^T \sum_{i=1}^n \hat{\lambda}_i \hat{v}_i \hat{v}_i^T \hat{v}_1 \\ &= \hat{v}_1^T \left(\hat{\lambda}_1 \hat{v}_1 \right) = \hat{\lambda}_1 (\hat{v}_1^T \hat{v}_1) = \hat{\lambda}_1 < 0 \end{aligned}$$

$$\text{For } M \in S_+^n: \quad \langle s, M \rangle = \langle \hat{v}_1 \hat{v}_1^T, M \rangle = \hat{v}_1^T M \hat{v}_1 \geq 0 \quad \text{b/c } M \in S_+^n$$

$$S_+^n \subseteq H_{s,b}^+$$

Copositive Matrices: a hard convex set

A ^{symmetric} matrix M is copositive if

$$x^T M x \geq 0 \quad \forall x \in \mathbb{R}_+^n.$$

① Q: Is the set of cop. matrices bigger or smaller than S_+^n ?

② Show that the set of copositive matrices is a convex cone.

FACT: In general it is intractable to determine whether a matrix M is copositive.